

Algebra II with Trigonometry

Chapter 5.1

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:

[4401267_alg-2trig-h-chp-5-1-note-docx.pdf](#)

Solutions

1. Find the least positive angle in degrees coterminal with

$$\frac{37\pi}{12} - 845^\circ.$$

First convert the radian measure to degrees:

$$\frac{37\pi}{12} \cdot \frac{180^\circ}{\pi} = 37 \cdot 15^\circ = 555^\circ.$$

So the angle is

$$555^\circ - 845^\circ = -290^\circ.$$

To find the least positive coterminal angle, add 360° :

$$-290^\circ + 360^\circ = 70^\circ.$$

$$\boxed{70^\circ}$$

2. A sector has perimeter 20 and central angle 2 radians. Find its area.

For a sector of radius r and angle θ , the perimeter is

$$2r + r\theta.$$

Here $\theta = 2$, so

$$2r + 2r = 4r = 20.$$

Thus

$$r = 5.$$

The area of a sector is

$$\frac{1}{2}r^2\theta.$$

Hence

$$\text{Area} = \frac{1}{2} \cdot 25 \cdot 2 = 25.$$

$$\boxed{25}$$

3. In a circle, the numerical value of the arc length of a sector equals the numerical value of the area of that sector. If the central angle is not 0, find it.

Let the radius be r and the central angle be θ radians.

The arc length is

$$s = r\theta,$$

and the sector area is

$$A = \frac{1}{2}r^2\theta.$$

We are told these numerical values are equal:

$$r\theta = \frac{1}{2}r^2\theta.$$

Since $\theta \neq 0$, divide by θ :

$$r = \frac{1}{2}r^2.$$

So

$$2r = r^2 \quad \Rightarrow \quad r(r - 2) = 0.$$

Since a circle cannot have radius 0, we get

$$r = 2.$$

The equation imposes no further restriction on θ besides $\theta \neq 0$. Thus the central angle can be **any nonzero radian measure**.

θ may be any nonzero real number (in radians)

4. A point moves on a circle of radius 6 cm. In the first half of a trip it turns through 150° ; in the second half it turns through $\frac{5\pi}{6}$ radians. Find the total distance traveled.

Distance traveled along a circle is arc length:

$$s = r\theta,$$

with θ in radians.

Convert 150° to radians:

$$150^\circ = \frac{5\pi}{6}.$$

So in the first half, the arc length is

$$6 \cdot \frac{5\pi}{6} = 5\pi.$$

In the second half, the angle is also $\frac{5\pi}{6}$, so the arc length is again

$$6 \cdot \frac{5\pi}{6} = 5\pi.$$

Therefore the total distance is

$$5\pi + 5\pi = 10\pi.$$

$$10\pi \text{ cm}$$

5. Two angles in standard position satisfy

$$\alpha + \beta = 7\pi, \quad \alpha - \beta = 250^\circ.$$

Find the least positive coterminal angle of α .

We first express everything in one unit. Convert 250° to radians:

$$250^\circ = \frac{250\pi}{180} = \frac{25\pi}{18}.$$

Now solve the system

$$\alpha + \beta = 7\pi, \quad \alpha - \beta = \frac{25\pi}{18}.$$

Add the equations:

$$2\alpha = 7\pi + \frac{25\pi}{18} = \frac{126\pi + 25\pi}{18} = \frac{151\pi}{18}.$$

Hence

$$\alpha = \frac{151\pi}{36}.$$

Now reduce modulo $2\pi = \frac{72\pi}{36}$:

$$\frac{151\pi}{36} - \frac{72\pi}{36} = \frac{79\pi}{36},$$

and again

$$\frac{79\pi}{36} - \frac{72\pi}{36} = \frac{7\pi}{36}.$$

This is positive and less than 2π , so it is the least positive coterminal angle.

$$\boxed{\frac{7\pi}{36}}$$

6. A wheel of radius r rolls without slipping. After exactly 9 revolutions, the distance traveled equals the area of a sector of a circle of radius r and angle θ radians. Find θ .

When a wheel of radius r rolls without slipping, one revolution covers one circumference:

$$2\pi r.$$

So after 9 revolutions, the distance traveled is

$$9(2\pi r) = 18\pi r.$$

The area of a sector of radius r and angle θ is

$$\frac{1}{2}r^2\theta.$$

Set these equal:

$$18\pi r = \frac{1}{2}r^2\theta.$$

Assuming $r > 0$, divide by r :

$$18\pi = \frac{1}{2}r\theta.$$

Thus

$$\theta = \frac{36\pi}{r}.$$

$$\boxed{\theta = \frac{36\pi}{r}}$$